Fall 2025

ECE 216A Homework 1

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Question 1

a.

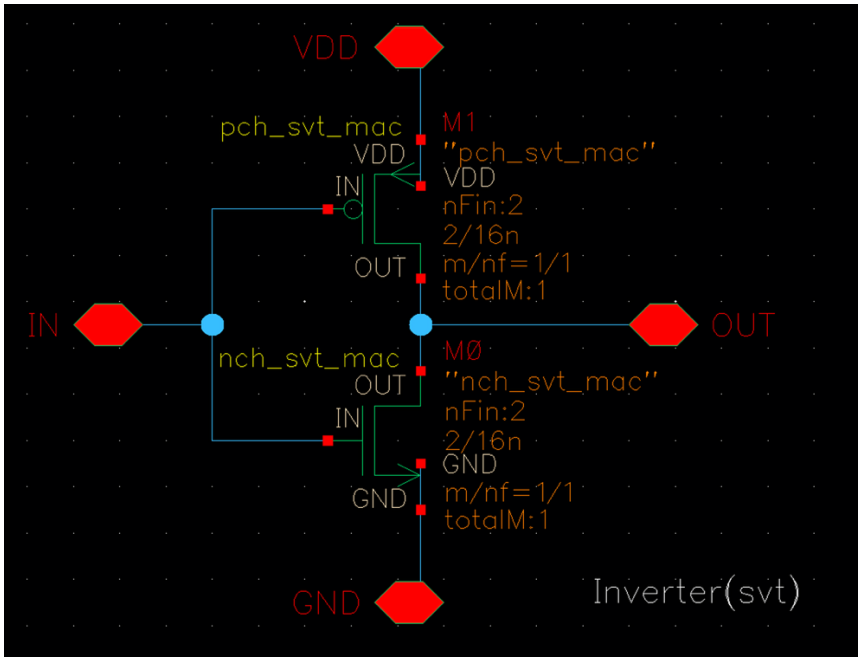


Figure 1. Schematic for CMOS Inverter with Standard Threshold Voltage

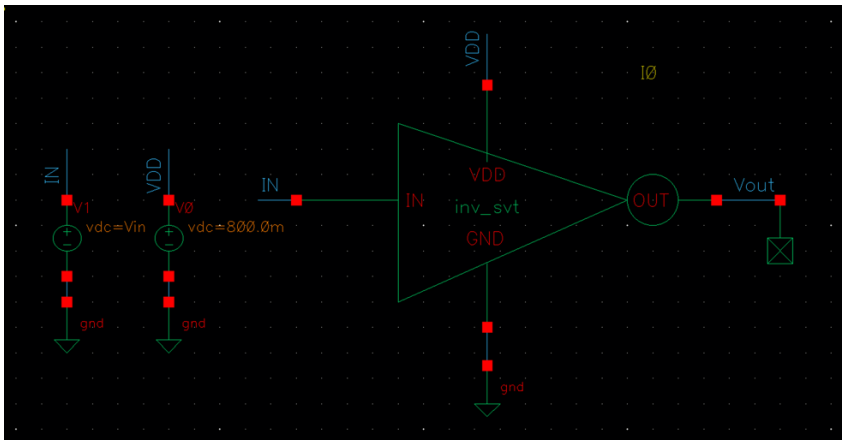


Figure 2. Testbench for CMOS Inverter with Standard Threshold Voltage

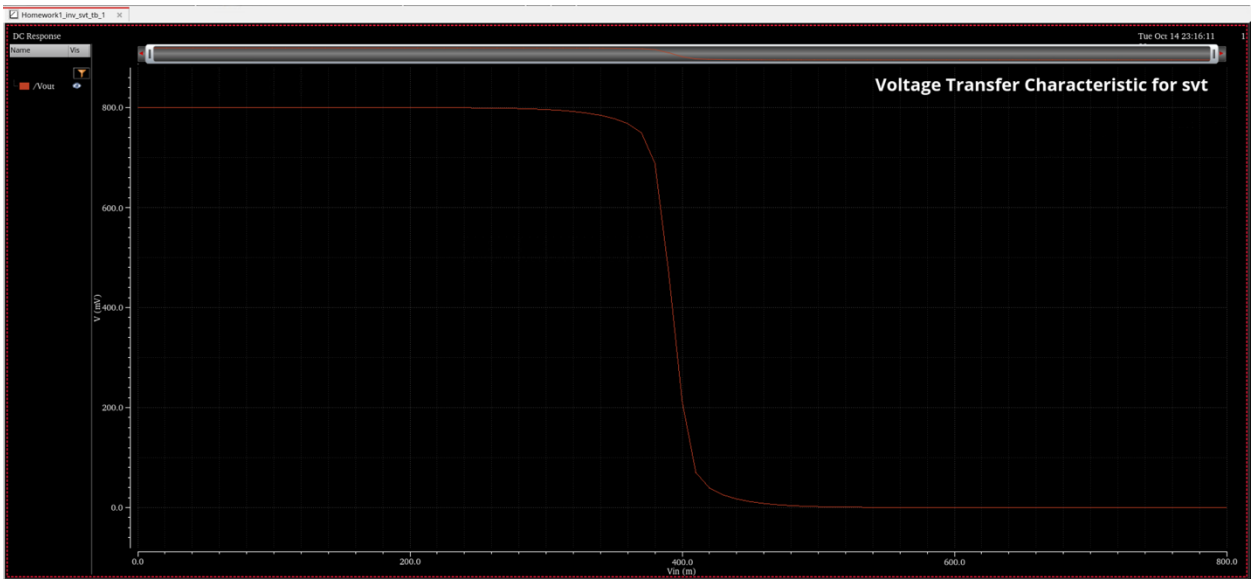


Figure 3. Voltage Transfer Characteristic for CMOS Inverter with Standard Threshold Voltage

Figure 3 is the simulation result of Voltage Transfer Characteristic for CMOS Inverter with Standard Threshold Voltage.

b.

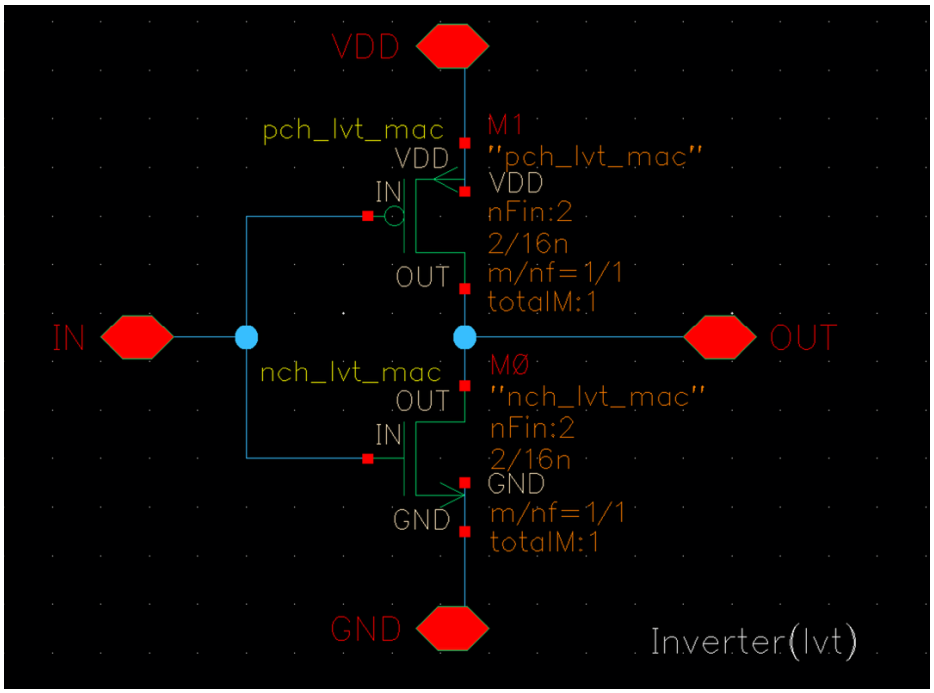


Figure 4. Schematic for CMOS Inverter with Low Threshold Voltage

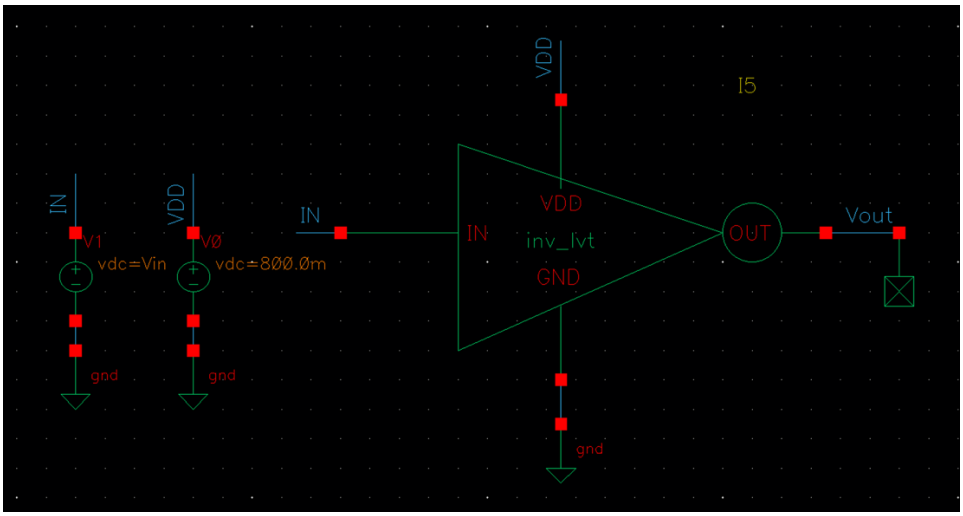


Figure 5. Testbench for CMOS Inverter with Low Threshold Voltage

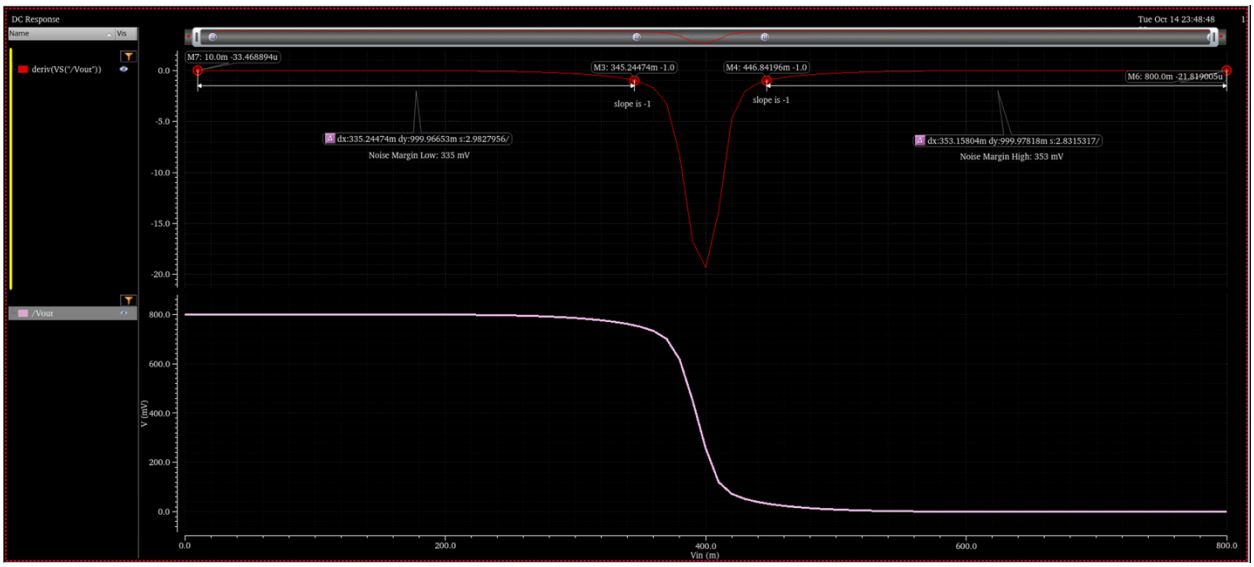


Figure 6. Voltage Transfer Characteristic for CMOS Inverter with Low Threshold Voltage (bottom) and its slope (top)

Figure 6 is the simulation result of Voltage Transfer Characteristic for CMOS Inverter with Low Threshold Voltage.

c.

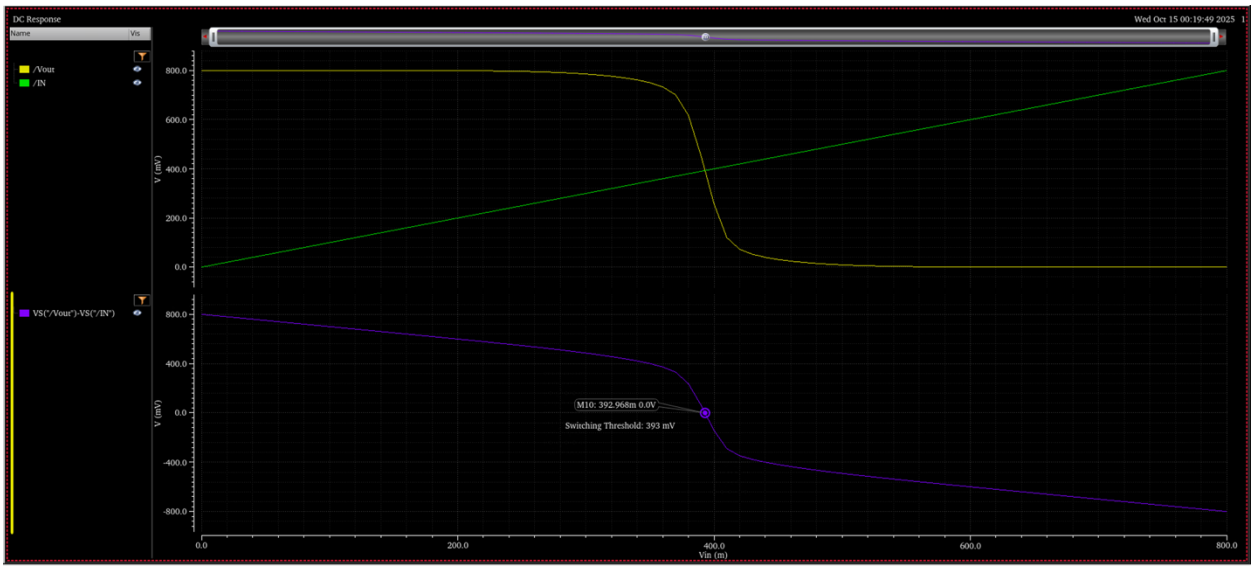


Figure 7. The difference between Vout and Vin as a Function of Vin from 0 to VDD (bottom)

Figure 6 shows that noise margin low is 335 mV and noise margin high is 353 mV. Also, Figure 7 shows that the switching threshold voltage (midpoint voltage) is 393 mV.

d.

Noise margin low and noise margin high are approximately equal and roughly symmetric around midpoint voltage. So, the pull-up strength and pull-down strength are nearly matched. Also, noise margins are large relative to the supply voltage of 0.8 V. Therefore, the CMOS inverter has large and balanced noise tolerance to ensure reliable logic level restoration.

Question 2

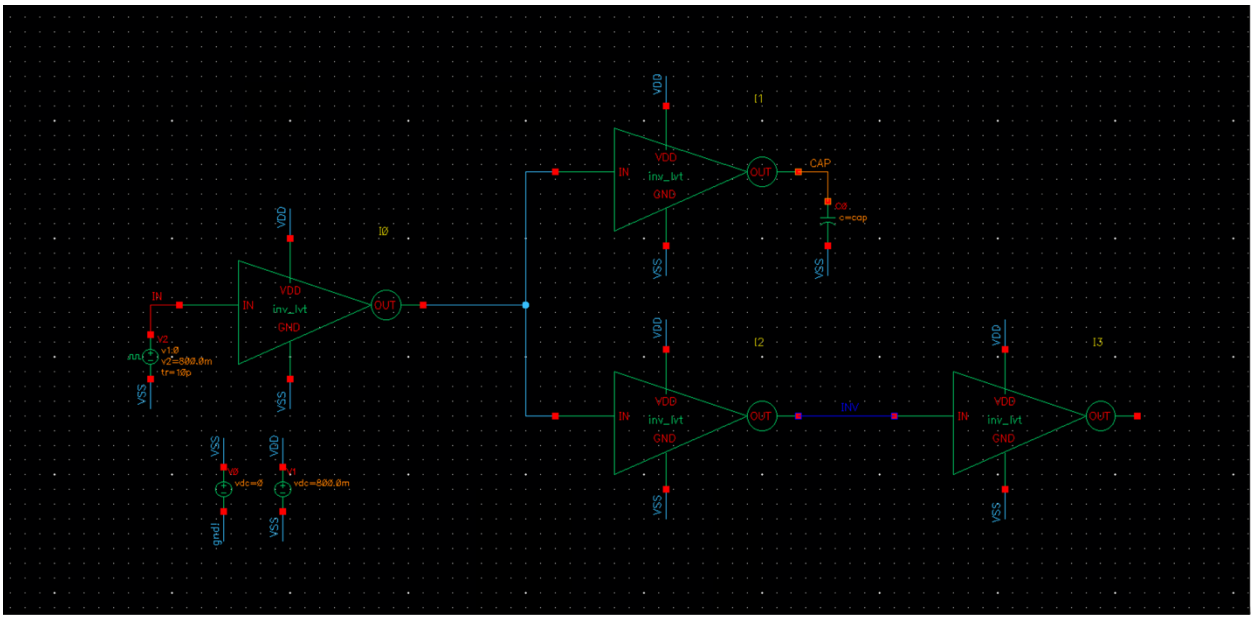


Figure 8. Testbench Setup to Measure Cgate

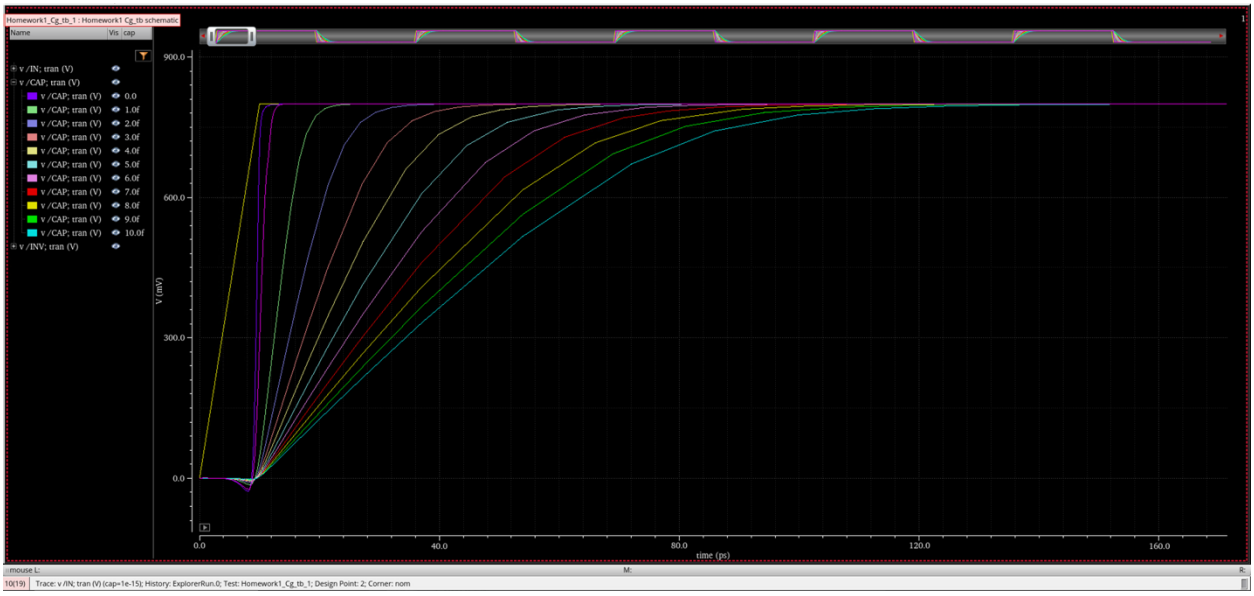


Figure 9. Voltage Change at Cap (Sweeping from 0 to 10fF) and Cgate Nodes After Applying a Vpluse at Input Node

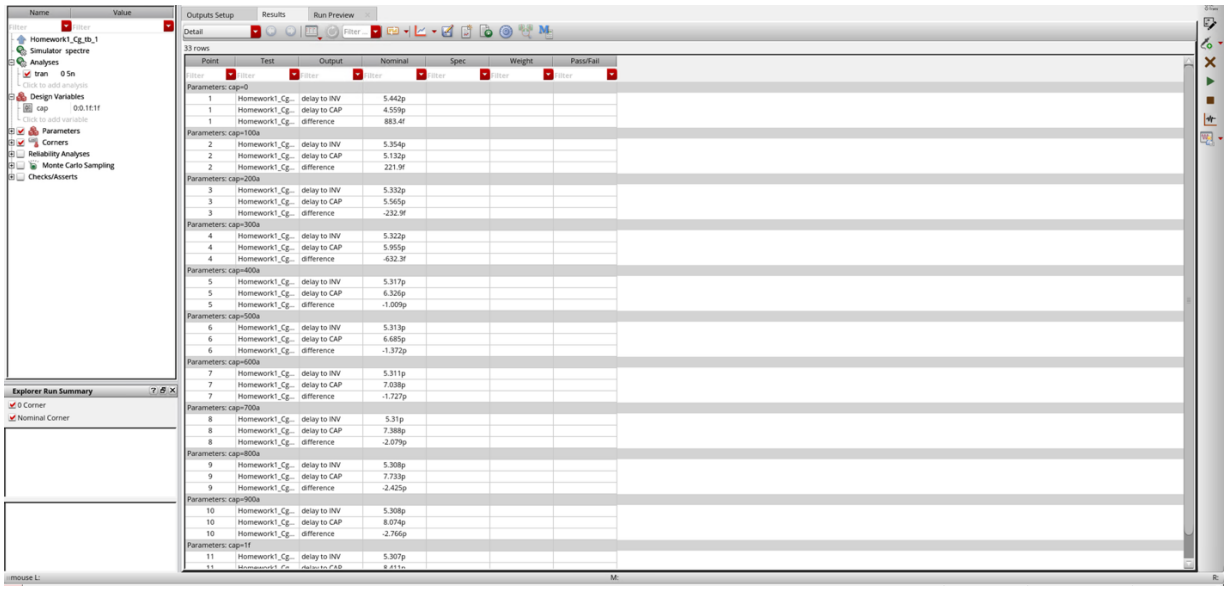


Figure 10. The Delay Difference Data Between Two Path When Sweeping Cap from 0 to 1fF

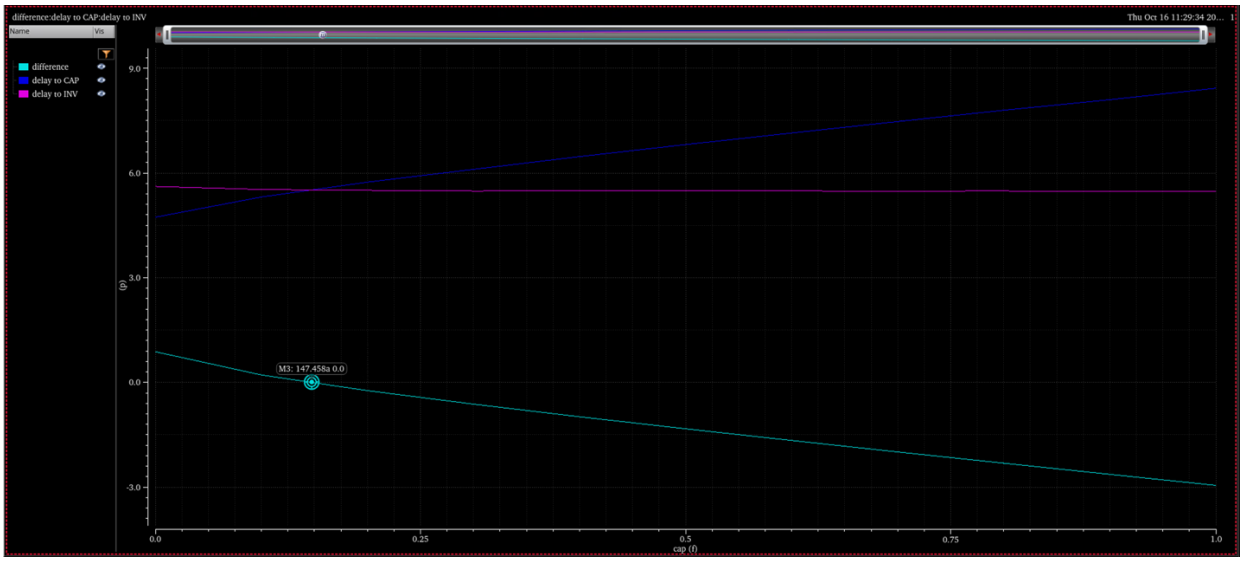


Figure 11. The Delay Difference Graph Between Two Path When Sweeping Cap from 0 to 1fF (bottom)

From Figure 11, the delay is same for two paths when Cap is 0.147 fF. So, Cgate is 0.147 fF.

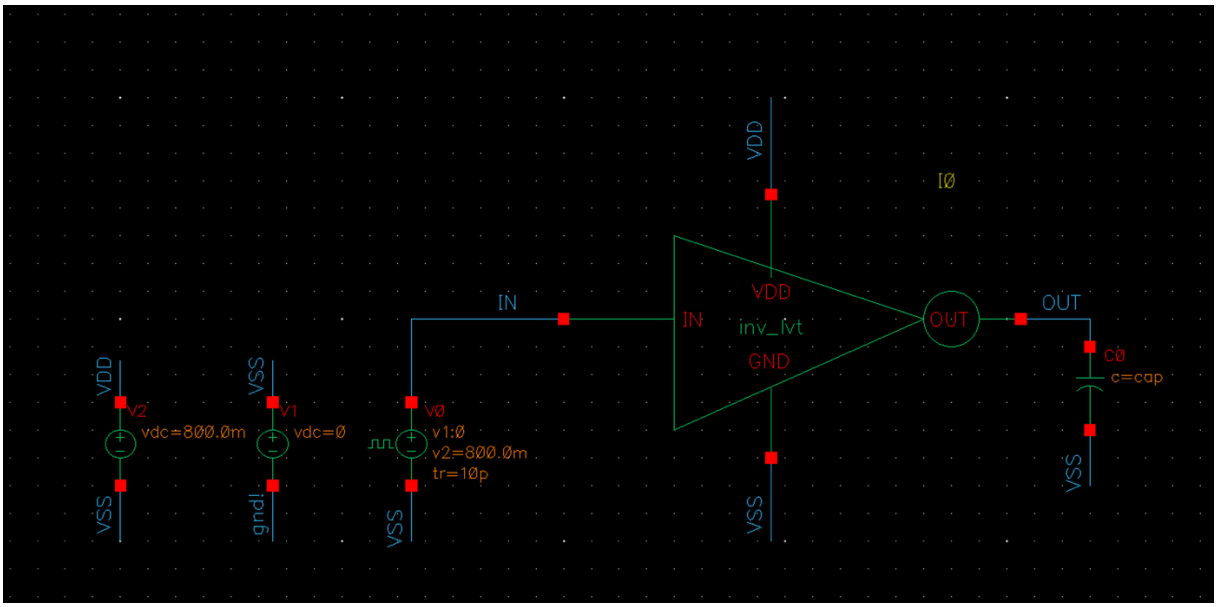


Figure 12. Testbench Setup to Measure Cpar and R

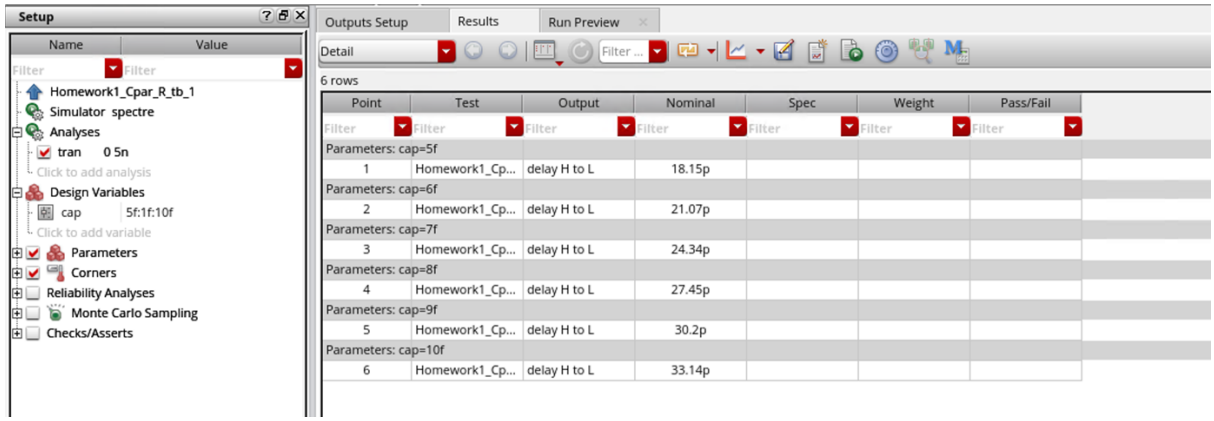


Figure 13. Propagation Delay Data for H to L when sweeping Cap from 5fF to10fF

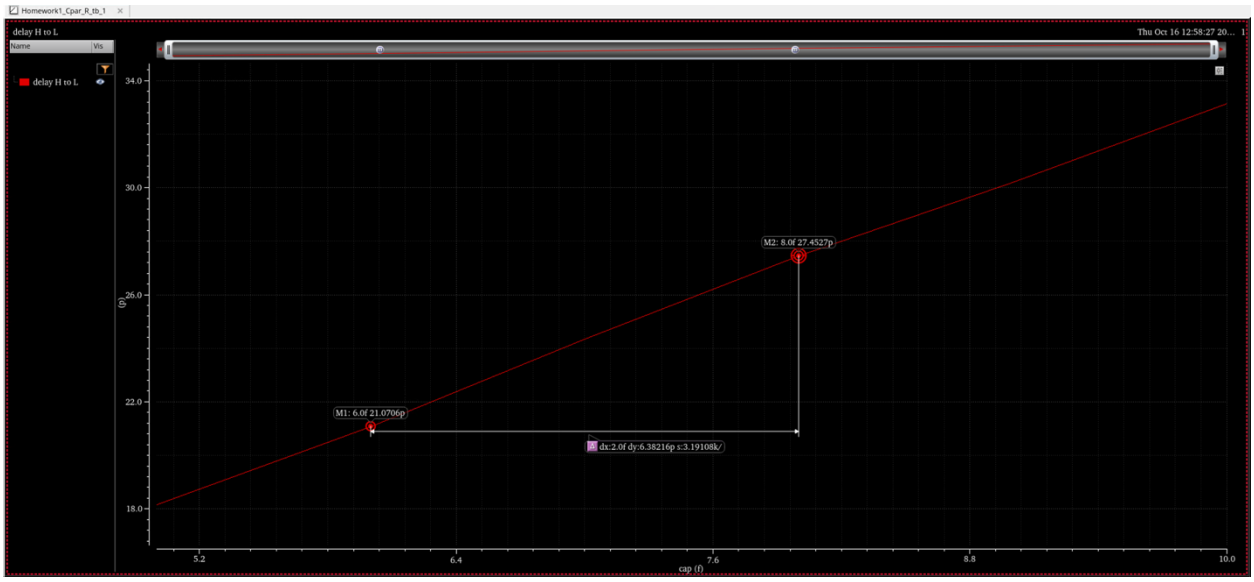


Figure 14. Propagation Delay Graph for H to L when sweeping Cap from 5fF to10fF

$$\text{Delay} = 0.69\,R\,(C_{par} + C_L)$$

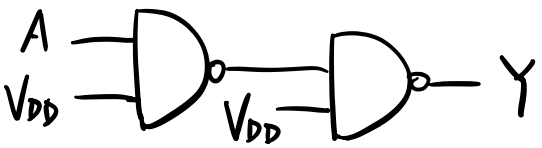
$$= 0.69\,R\,C_L + 0.69\,R\,C_{par}$$

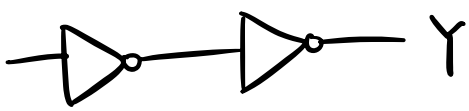
$$\begin{cases} C_L = 6\,fF, \ \text{Delay} = 21.07\,ps \\ C_L = 8\,fF, \ \text{Delay} = 27.45\,ps \end{cases}$$

$$\Rightarrow \begin{cases} C_{par} = 0.6\,fF \\ R = 4.625\,k\Omega \end{cases}$$

In Summary, Cgate is 0.15 fF, Cpar is 0.6 fF, and R is 4.625 kΩ.

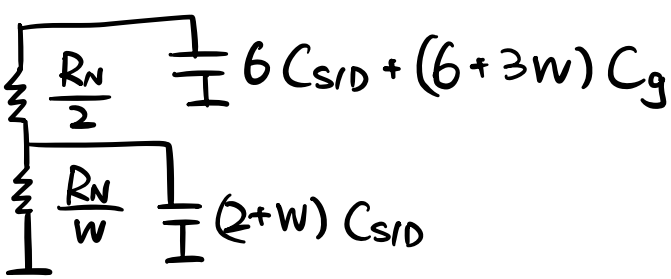
3.

(a) ⇒ 

⇒ 

⇒ $\boxed{Y = A \quad (\text{buffer})}$

(b) A ⌐¯ → B ¯⌐



$$\tau = [6C_{SID} + (6+3W)C_g]\left(\frac{R_N}{2} + \frac{R_N}{W}\right) + (2+W)C_{SID}\frac{R_N}{W}$$

$$= [3C_o + (6+3W)C_o]\left(\frac{R_o}{2} + \frac{R_o}{W}\right) + \left(1+\frac{W}{2}\right)C_o\frac{R_o}{W}$$

$$= \frac{(3W+10)(W+2)}{2W} R_o C_o$$

$$= \frac{3W^2+16W+20}{2W} R_o C_o \quad fF$$

$$\frac{d\tau}{dw} = \frac{R_0 C_0}{2} \cdot \frac{w(6w+16) - (3w^2+16w+20)}{w^2} = 0$$

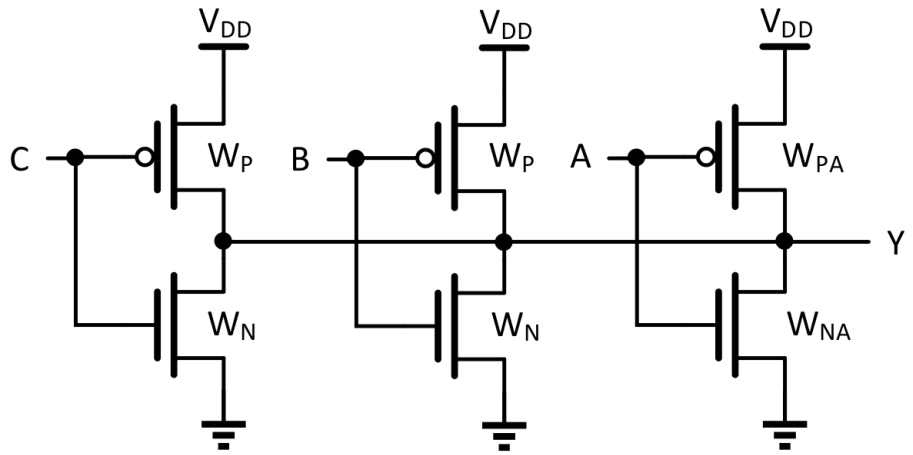
$$W_{opt} = \sqrt{\frac{20}{3}} = 2.58 \approx 3$$

(c) If it applies diffusion sharing, C_{par} at B and C_{int} between 2 NMOS both decreases. As the total load decreases, bottom NMOS drives less load, which can be sized small.

So, W_{opt} decreases.

4.

a.



$$\beta = \frac{\mu_n}{\mu_p} = 3 = \frac{R_P'}{R_N'}$$

assume $R_N' = R'$, then $R_P' = 3R'$

$$R_{PC} = R_{PB} = \frac{R_P'}{W_P} = \frac{3R'}{W_P}$$

$$R_{PA} = \frac{R_P'}{W_{PA}} = \frac{3R'}{W_{PA}}$$

$$R_{NC} = R_{NB} = \frac{R_N'}{W_N} = \frac{R'}{W_N}$$

$$R_{NA} = \frac{R_N'}{W_{NA}} = \frac{R'}{3}$$

The output is considered logic 1 if voltage at that node is greater than $\frac{V_{DD}}{2}$ and considered logic 0 if voltage at that node is less than $\frac{V_{DD}}{2}$.

To function correctly, the output is logic 0 if pull-up resistance is larger than pull-down resistance; the ouput is logic 1 if pull-down resistance is larger than pull up.

AOI Gate $Y = A + \overline{B \cdot C}$

A	B	C	Y	
0	0	0	1	
0	0	1	1	$(R_{PA} \parallel R_{PB}) < R_{NC}$
0	1	0	1	$(R_{PA} \parallel R_{PC}) < R_{NB}$
0	1	1	0	$(R_{NB} \parallel R_{NC}) < R_{PA}$
1	0	0	0	$R_{NA} < (R_{PB} \parallel R_{PC})$
1	0	1	0	$(R_{NA} \parallel R_{NC}) < R_{PB}$
1	1	0	0	$(R_{NA} \parallel R_{NB}) < R_{PC}$
1	1	1	0	

$$(R_{PA} \parallel R_{PB}) < R_{NC} \qquad \left(\frac{3R'}{W_{PA}} \parallel \frac{3R'}{W_P}\right) < \frac{R'}{W_N}$$

$$(R_{PA} \parallel R_{PC}) < R_{NB} \qquad \left(\frac{3R'}{W_{PA}} \parallel \frac{3R'}{W_P}\right) < \frac{R'}{W_N}$$

$\Big\}$ ①

$$(R_{NB} \parallel R_{NC}) < R_{PA} \qquad \left(\frac{R'}{W_N} \parallel \frac{R'}{W_N}\right) < \frac{3R'}{W_{PA}} \quad ②$$

$$R_{NA} < (R_{PB} \parallel R_{PC}) \qquad \frac{R'}{3} < \left(\frac{3R'}{W_P} \parallel \frac{3R'}{W_P}\right) \quad ③$$

$$(R_{NA} \parallel R_{NC}) < R_{PB} \qquad \left(\frac{R'}{3} \parallel \frac{R'}{W_N}\right) < \frac{3R'}{W_P}$$

$$(R_{NA} \parallel R_{NB}) < R_{PC} \qquad \left(\frac{R'}{3} \parallel \frac{R'}{W_N}\right) < \frac{3R'}{W_P}$$

$\Big\}$ ④

① $\Rightarrow$ $3W_N < W_{PA} + W_P$

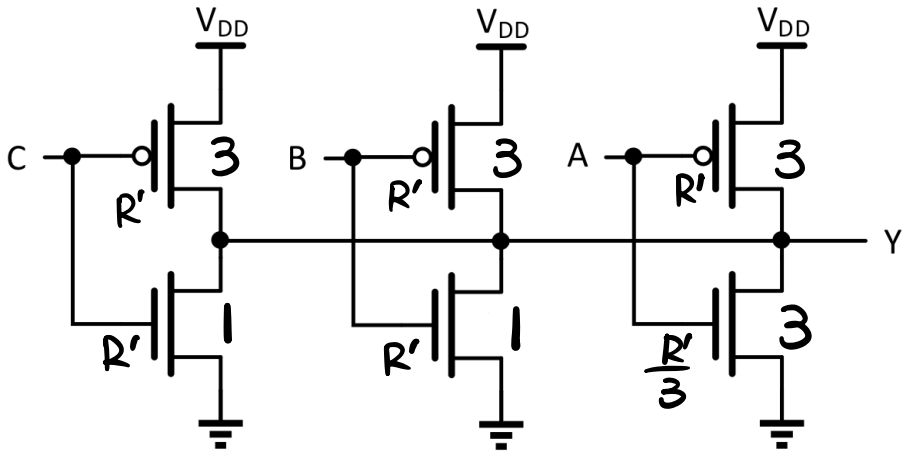
② $\Rightarrow$ $W_{PA} < 6W_N$

③ $\Rightarrow$ $W_P < 4.5$

④ $\Rightarrow$ $W_P < 9 + 3W_N$

$$\Rightarrow \begin{cases} 3W_N < W_{PA} + W_P \\ W_{PA} < 6W_N \\ W_P < 4.5 \end{cases}$$

b.



worst-case logical effort is at a node when its has largest C_{gate} and largest equivalent R.

node A when A = 1, B = 0, C = 0

$$I_{PUN} \propto \frac{1}{R' \| R'} = \frac{2}{R'}$$

$$I_{PDN} \propto \frac{1}{R' / 3} = \frac{3}{R'}$$

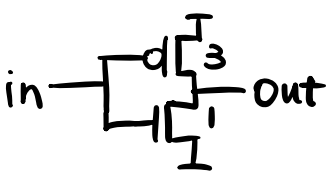
$$I_{out} = I_{PDN} - I_{PUN}$$

$$\propto \frac{3}{R'} - \frac{2}{R'}$$

$$\propto \frac{1}{R'}$$

$$\Rightarrow R_{gate} = R'$$

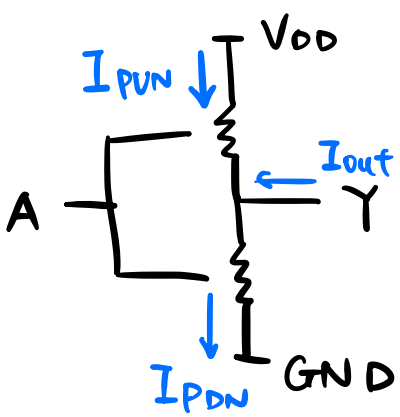
reference inv:



$C_{inv} = 4C'_{gate}$

$R_{inv} = R'$

$R_p' = 3R_N' = 3R'$



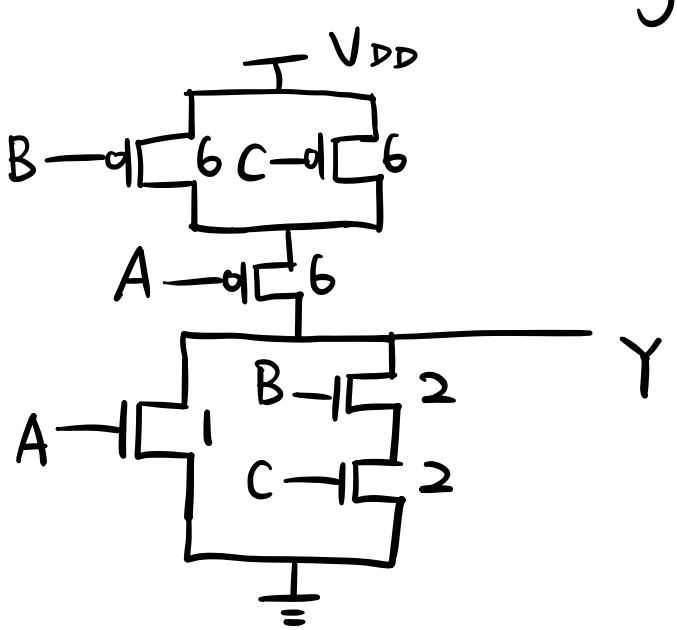
$$g = \frac{R_{gate} C_{gate}}{R_{inv} C_{inv}} = \frac{R' \cdot 6C'_{gate}}{R' \cdot 4C'_{gate}} = \boxed{\frac{3}{2}}$$ g

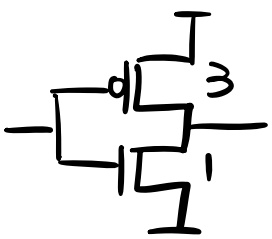
worst-case parasitic delay happens at the node which has the largest PUN or PDN equivalent R, same as above at node A.

$$\gamma p = \gamma \frac{R_{gate} C_{gate,self}}{R_{inv} C_{inv,self}} = \gamma \frac{R' \cdot 14 C'_{diff}}{R' \cdot 4 C'_{diff}} = \boxed{\frac{7}{2}\gamma}$$

p

C. static CMOS realizing $Y = \overline{A + B \cdot C}$





worse-case : $g = \frac{8 C'_{gate}}{4 C'_{gate}} = \boxed{2}_g$

$\gamma_P = \gamma \frac{9 C'_{diff}}{4 C'_{diff}} = \boxed{\frac{9}{4}\gamma}_P$

Ganged CMOS has smaller logical effort but larger parasitic delay compared to static CMOS.

The benefits are less input load, better for high fan-out path, reducing stages compared to static CMOS.

5.

$$\tau_{elmore} = C_8(R_1 + R_3 + R_6 + R_8) + C_6(R_1 + R_3 + R_6)$$
$$+ C_5(R_1 + R_3) + C_3(R_1 + R_3) + C_7(R_1 + R_3)$$
$$+ C_4 R_1 + C_2 R_1 + C_1 R_1$$

$$= (4 + 3 + 2 + 2 + 2 + 1 + 1 + 1)\,RC$$

$$= 16\,RC$$

$$= 16 \times 10 \times 100f$$

$$= \boxed{16\,ps}$$

$$\text{Delay} = 0.69\,R\,(C_{par} + C_L)$$

$$= 0.69\,R\,C_L + 0.69\,R\,C_{par}$$

$$\begin{cases} C_L = 6\,fF, \ \text{Delay} = 21.07\,ps \\ C_L = 8\,fF, \ \text{Delay} = 27.45\,ps \end{cases}$$

$$\Rightarrow \begin{cases} C_{par} = 0.6\,fF \\ R = 4.625\,K\Omega \end{cases}$$